MTH2005 — Practical 2: ODEs

LEARNING OBJECTIVES

1. Implement forward-Euler and leapfrog methods in Python, and practice plotting results.
2. Apply and text theory regarding stability of ODE methods that was covered in lectures.

INTRODUCTION

Welcome to the second practical session of MTH2005, Mathematical Modelling: Theory and Practice. If you'd like a reminder of how to open and run Python, please see the sheet for practical 1. Python code solutions to the practicals will be posted on ELE, so please spend some time checking your answers against these. Your solutions to these practicals will be an important resource when it comes to completing the coursework.

TASK

Today we look at the key problem

$$\frac{\mathrm{d}x}{\mathrm{d}t} = F(x)\,,$$

studied in the lectures. The Python code in the accomapanying script (`practical2.py`) solves the above ordinary differential equation (ODE) using both forward-Euler and leapfrog methods. By commenting or uncommenting the code there are options of *exponential decay* where

$$F(x) = -\lambda x\,,$$

and $\lambda > 0$, or for *oscillations* where

$$F(x) = i\omega x\,,$$

and $\omega > 0$. The aim of this practical is to take this code apart and understand its components. It will be helpful to refer to your lecture notes during the practical.

Investigations:

1. Spend some time familiarising yourself with the code. Change to the expoential decay case, and consider just the forward-Euler solution by commenting out the leapfrog solution and the code for plotting it. By varying λ and keeping Δt constant, explore the range of values of $\lambda\Delta t$ derived for stability and instability in the lectures. Do your numerical resuts agree with the theory?
2. Now change to the oscillation case. How do the stability properties of the forward-Euler and leapfrog schemes change? Compare your results with the lecture notes.
3. Add a third scheme: the backwards-Euler (or implicit-Euler) scheme. How do the results change?